Samuele Scaglioni

samueles@uchicago.edu | Personal website | GitHub | LinkedIn

Education

University of Chicago — Chicago, IL
Master of Arts in Economics; Dean's Scholarship — *Expected June 2027*

University of Illinois Chicago — Chicago, IL
M.S. Industrial Engineering; GPA: 4.0/4.0 (top 1%); 50% tuition waiver — *December 2024*

Politecnico di Milano — Milan, Italy
M.S. Management and Production Engineering; 110/110 cum laude*, GPA: 29/30 (top 1%)* — *July 2024*
B.S. Management and Production Engineering — *July 2022*

Relevant coursework: Mathematical Analysis I–II (proof-based), Linear Algebra, Probability and Statistics, Stochastic Processes, Data Science (PhD), Operations Research and Game Theory, Development Economics, Computer Science.

Research Experience

Politecnico di Milano — Milan, Italy
Research Assistant to Prof. Daniele Siena, Macroeconomics — *Nov 2024 – Nov 2025*

- Processed daily euro-area yield-curve data and extracted functional principal components to summarize movements across maturities around European Central Bank policy announcements.
- Implemented a proxy structural vector autoregression (SVAR) to distinguish conventional from unconventional monetary-policy shocks and support the project's empirical identification strategy.

University of Illinois Chicago — Chicago, IL
Research Assistant to Prof. Hadis Anahideh, Econometrics and Data Science — *Jan 2024 – Jun 2024*

- Built a reproducible Python pipeline combining random forests and causal forests to estimate conditional average treatment effects and predict counterfactual outcomes.
- Designed budget-constrained treatment-assignment rules for policy targeting; findings presented at INFORMS 2024.

Other Professional Experience

European Central Bank — Frankfurt, Germany
Analyst — *Oct 2025 – Present*
Stress Test Modelling Trainee — *Oct 2024 – Sep 2025*

- Compared econometric and machine-learning models for bank-level probability of default and translate micro-level predictions into stress-test applications; research presented at EcoMod 2026 and CEF 2026.
- Developed a Hill-estimator tail-risk indicator with statistically significant predictive content at 12–24-month horizons, informing validation work supporting the 2025 EU-wide stress test.
- Built a MATLAB pipeline and interactive dashboard for model-questionnaire analysis across 80+ euro-area banks, reducing review time by approximately 50%.
- Support event-study research on tariff announcements and abnormal equity returns using generalized autoregressive conditional heteroskedasticity (GARCH) models; contribute to artificial-intelligence tools for supervisory analytics.

Publications & Working Papers

- **Learning Probability of Default and Stress Testing** — Econometric and machine-learning models for micro-level credit-risk stress testing; with Luca Nocciola (ECB). *ECB Working Paper Series No. 3277*; presented at EcoMod 2026 and CEF 2026.
- **Institutional Cleansing and Political Scars** — Mafia-related municipal dissolution, political participation and supply, and local credit disruption; with Nordine Abidi (IMF) and Ixart Miquel-Flores (ECB). *Work in progress.*
- **IMPACT: Robust Causal Treatment Analysis under Resource Constraints** — Budget-aware treatment-effect estimation and counterfactual targeting; with Hadis Anahideh (UIC). *Submitted to EAAMO 2026.*

Technical Skills and Additional Information

Programming: Python, MATLAB, Stata, R, SQL.
Tools: Git, LaTeX, Excel.
Methods: causal inference, event studies, time-series econometrics, machine learning and deep learning, stress testing.
Research interests: financial stability, macro-finance, political economy, applied econometrics.
Languages: Italian (native), English (C2; IELTS 8.5).
GRE: 165 Quantitative, 165 Verbal, 4 Analytical Writing.